

Programme	B. Sc. Computer Science				
Course Title	Data Mining				
Type of Course	Major				
Semester	V				
Academic Level	300 – 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	1. Basics of statistics				
Course Summary	This course provides an introduction to the principles, techniques, and applications of data mining.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the fundamental concepts and principles of data mining.	U	C	Instructor-created exams / Quiz
CO2	Demonstrate proficiency in preprocessing techniques such as cleaning, transformation, and reduction of data.	U	P	Assignment / Seminar presentations/ Exams
CO3	Understand popular data mining algorithms and models, such as decision trees, k-means clustering, and association rule algorithms.	U	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Explore various methods to Evaluate and interpret the results of data mining models using appropriate performance metrics.	U	C	Instructor-created exams / Home Assignments
CO5	Understand the role of data mining in extracting patterns and knowledge from large datasets.	U	P	Writing assignments/ exams/ Seminar
CO6	Apply data mining techniques to real-world problems and datasets, emphasizing practical applications in various domains	Ap	P	Case Study
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Introduction to Data Mining		10	15
	1	Introduction- Data mining defining, KDD vs Data mining, DBMS vs data mining	2	
	2	What kind of data can be mined? - database data, data warehouse, transactional data, other types	2	
	3	What kind of patterns can be mined? - Class/Concept Description: Characterization and Discrimination, Mining Frequent Patterns, Associations, and Correlations, Classification and Regression for Predictive Analysis, cluster analysis, outlier analysis	3	
	4	Technologies used- statistics, machine learning, data base systems and ware house, information retrieval (Introduction only)	3	
II	Data Preprocessing		14	20
	5	Data Preprocessing: An Overview	2	
	6	Data Cleaning- missing value, noisy data, Data Cleaning as a Process	2	
	7	Data Integration- Entity Identification Problem, Redundancy and Correlation Analysis, Tuple Duplication, Data Value Conflict Detection and Resolution	3	
	8	Data Reduction - Wavelet Transforms, Principal Components Analysis, Attribute Subset Selection, Regression and Log-Linear Models: Parametric Data Reduction, Histograms,	4	
	9	Data Transformation and and Data Discretization- Data Transformation by Normalization, Discretization by Binning	3	
III	Association Rule Mining & Classification		10	15
	10	Introduction to Association Rule Mining Frequent Itemset, Closed Itemset, and Association Rules	1	
	11	Frequent Itemset Mining Apriori Algorithm, Generating Association Rules from Frequent Itemsets	1	
	12	Introductio to classification: Decision tree	2	
	13	Attribute Selection measures in decision tree	2	
	14	Bayes Classification methods	2	
	15	Techniques to Improve Classification Accuracy	2	
IV	Clustering, Outlier detection		14	20
	16	Introduction to unsupervised techniques: challenges	2	
	17	Clustering- K Means	2	
	18	Variants of k- Means	2	

	19	Hierarchical clustering	2	
	20	Density Based clustering- DBScan	2	
	21	Introduction to outliers and novelty detection	2	
	22	Recommender system	2	
V	Open Ended Module: Case Studies		12	
	1	• Real-world applications of data mining • Case studies and projects • Ethical considerations in data mining		

References

- "Han, J., Kamber, M., & Pei, J. (2011). Data mining: Concepts and techniques. Morgan Kaufmann."
- Data Mining Techniques - Arun K. Pujari
- Jiawei Han and Micheline Kamber, Data Mining Concepts & Techniques, Second Edition, Elsevier.
- Pang Ning Tan, Michael Steinbach and Vipin Kumar, Introduction To Data Mining, Pearson Education, 2007.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	-	1						
CO 2	1	-	-	-	1	1						
CO 3	1	-	2	-	2	2						
CO 4	1	-	1	-	1	1						
CO 5	1	-	1	-	1	1						
CO 6	-	-	1	1	2	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar

- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4	✓	✓	✓
CO 5	✓	✓	✓
CO 6	✓	✓	